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Earthwork Network Architecture (ENA): Research for Earthwork Quantity Estimation Method Improvement with Large Language Model

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Abstract: Accurate earthwork quantity estimation is essential for effective project planning and cost management in the Architecture, Engineering, and Construction (AEC) industry. Traditional methods for quantity takeoff are often time-consuming and susceptible to human error, particularly when working with unstructured datasets such as CAD drawings. This study introduces the Earthwork Network Architecture (ENA), a novel deep learning framework that incorporates Large Language Models (LLMs), Multi-Layer Perceptron (MLP), Long Short-Term Memory (LSTM) networks, and Transformers to automate and enhance the accuracy of earthwork quantity estimation. We assume that if LLMs can be trained effectively using such unstructured construction dataset, the effects such as improved accuracy and the challenges of LLMs can be clearly examined. Among the architectures tested, the LLM-based ENA demonstrated superior performance, achieving faster convergence, greater loss reduction, and higher classification accuracy, with a Quantity Takeoff Classification accuracy of 97.17%. However, the LLMs required significantly more computational resources compared with other models. These findings suggest that LLMs, typically used in natural language processing, can be effectively adapted for complex AEC datasets. This study lays the groundwork for future AI-driven solutions in the AEC industry, underscoring the potential of LLMs and Transformers to automate the quantity takeoff process and manage multimodal data in construction projects.

Keywords: earthwork quantity estimation; large language models (LLMs); deep learning in construction; transformer-based models; automated quantity takeoff



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1. Introduction

Accurately estimating earthwork quantities is a crucial aspect of project planning and cost estimation in civil and construction projects. Earthwork, which involves excavation and filling, often represents a significant portion of construction costs, making precise estimation vital for the success of these projects. The accuracy of these estimates directly affects key elements such as project budgeting, scheduling, and resource management [1]. Inaccurate estimates can result in either underestimation or overestimation of material volumes, both of which have serious financial and operational consequences [2]. Underestimating the required volumes can lead to unexpected cost overruns, delays, and the need for additional resources, ultimately disrupting the project timeline and straining relationships with stakeholders. Conversely, overestimating volumes can inflate project costs, resulting in wasted materials and inefficiencies in labor and equipment utilization. Furthermore, precise earthwork estimation is essential for addressing environmental considerations, such as proper drainage and slope stability, which helps prevent future structural issues and ensures compliance with regulatory standards [3]. Therefore, accurate earthwork quantity

estimation is indispensable for maintaining the financial viability, operational efficiency, and environmental sustainability of construction projects.

Estimating earthwork quantities is typically carried out by estimators before the start of construction projects. However, despite their expertise, this process is highly dependent on the skill and judgment of engineers and surveyors, making it particularly prone to frequent human errors [4]. This vulnerability is even more evident in large-scale projects [5], where the sheer volume and complexity of data exponentially increase the likelihood of mistakes. Human errors in interpreting design drawings, misreading topographical contours, or performing manual volume calculations can occur at multiple stages, often leading to inaccurate estimates [6–8]. The frequent occurrence of such errors highlights the critical need for automated earthwork quantity estimation systems. By leveraging advanced technologies such as deep learning and CAD integration, automation can greatly minimize human error, improve accuracy, and enhance efficiency [9]. Automating these calculations ensures more reliable estimates, protects project profitability, optimizes resource management, and helps maintain project schedules, underscoring the essential role of automation in modern construction practices.

This paper presents a novel approach called the Earthwork Network Architecture (ENA), which utilizes deep learning to optimize and automate the quantity takeoff process for road cross-sections (x-sections). The proposed algorithm transforms data from Computer-Aided Design (CAD) drawings into a structured and trainable dataset, facilitating the extraction of critical features for effective training and evaluation of the ENA model. This paper develops and evaluates the ENA by (1) employing state-of-the-art deep learning algorithms identified through an extensive literature review and (2) implementing the proposed ENA into a real-world road construction project in Korea as a case study.

The proposed ENA extracts data from Computer-Aided Design (CAD) drawings and converts them into structured, trainable datasets. The ENA applies various deep learning models, particularly Large Language Models (LLMs) and Transformers, which offer significant advantages in processing complex CAD data and delivering precise estimates. The ENA then classifies earthwork quantities of civil and construction projects using the topological connectivity algorithm by creating a closed polyline in cross-section construction drawings [10]. The ENA framework is trained using these models, with features such as cut, fill, and other earthwork-related parameters extracted from the extracted datasets. The comparative analysis of different models, including the MLP, LSTM, and Transformers, revealed that LLMs outperformed others in terms of accuracy and efficiency, especially for large-scale projects.

This study demonstrates the effectiveness of the Earthwork Network Architecture (ENA) in automating and improving the accuracy of earthwork volume estimation through a case study. To develop the ENA model, the drawing vector dataset was converted into an LLM training data format. The performance of the proposed ENA model was compared with deep learning models with different architectures. In reference, this study highlights the challenges and limitations of traditional manual estimation processes and showcases how the ENA framework mitigates these issues by streamlining the extraction of features from CAD drawings. The key contributions of this research include the development of a novel ENA framework, the application of advanced deep learning architectures to earthwork quantity estimation, and the demonstration of how automated methods can significantly reduce manual labor and errors. This study offers practical insights for civil engineering professionals, paving the way for the future adoption of AI-driven tools in project planning and cost estimation. It also sets the groundwork for expanding the use of deep learning in other construction management tasks, contributing to the growing body of knowledge on AI applications in civil engineering.

2. Related Work

The application of deep learning and AI in automated earthwork quantity estimation is still evolving, showing promising potential to automate complex tasks. While significant

research has explored AI's role in improving construction productivity (e.g., equipment tracking, crack detection, and 3D point cloud enhancement [11]), challenges such as limited structured datasets, model interpretability, and autonomous decision-making persist. Current implementations in construction machinery and robotics have demonstrated enhanced perception, navigation, and control [12]. However, the application of deep learning to automate earthwork quantity takeoff remains underdeveloped.

Some studies, like vision systems for tracking equipment and analyzing posture, have shown promise in enhancing efficiency [13,14]. Similarly, object detection models have been applied to boost project efficiency [15]. However, issues like data variability and lack of structured collection hinder widespread deployment. Emerging tools like Vision Transformers (ViT) facilitate real-time monitoring and advanced design simulations in the Architecture, Engineering, and Construction (AEC) sector [16], highlighting the need for focused research on automating estimation processes.

AI-based machine guidance (MG) and control systems have led to significant productivity gains in trench excavation [17]. While models such as CNN-based object detection for excavator tasks [18] and DNN-based productivity benchmarking [19] provide real-time insights, they have yet to fully address the complexities of automated quantity estimation. Moreover, powerline detection using UAV-based LiDAR data [20,21] and road extraction models like Seg-Road [22] suggest potential for adapting these techniques to automate earthwork quantity calculations. Yet, successful adoption in earthwork applications remains limited due to the need for high-quality input data.

Deep learning has shown promise in various earthwork-related tasks, such as tunnel point cloud segmentation [23] and BIM-GIS integration for real-time spatial information [24,25]. Technologies like K-Net have further potential in material classification for real-time earthwork estimation [26]. Additionally, graph-based learning and automated CAD analysis have been explored to capture geometric features for architectural design [9,27], and similar approaches could enhance earthwork estimation automation.

Simulation models like the LSTM-Transformer fusion have been applied for transit flow prediction [28], while reinforcement learning has been explored for optimizing cut and fill operations [29]. However, these models are not yet widely adopted for earthwork automation. In construction management, AI has been used to predict vehicle speeds in earthwork simulations using improved XGBoost models [30], and progress monitoring systems integrated with UAV-based image analysis have shown potential despite challenges in complex data segmentation [3,31,32]. GPT models have been used to predict construction accidents, contributing to safer and more efficient operations [33,34]. Combining domain knowledge with AI, such as in subway construction risk management [35] and 3D modeling of earthwork sites using deep learning [32,36–39], underscores AI's expanding role in optimizing construction management and quantity estimation.

It can be seen that most of the deep learning models applied in the AEC field investigated above are the MLP, CNN, U-NET, LSTM, etc., and the emerging LLM-based research is lacking. Especially, the Transformer-based models such as LLM are not widely used, which shows that it is not easy to build learning data related to LLM in the field of construction engineering. To utilize LLM in a specific construction field, it is necessary to develop learning data from a more specialized and engineering perspective rather than using cases such as translation, writing, summarization, and proposal, in which LLM is active. As the survey results show, these methods are not yet widely publicized or active in the construction field. These challenges appear to stem from inherent features of the construction sector, as detailed in the literature review. Key issues include stringent security regulations, a conservative stance driven by high-risk environments, and the large volumes of unstructured data.

In this study, if we learned how to extract the earthwork quantity from the drawing using LLMs, we assumed that we would achieve better performance than other deep learning models. Even if the LLM-based ENA model performs better as assumed, it will not be cost-effective. By evaluating the performance between LLMs and other deep learning

model architectures, we can confirm the advantages and disadvantages of the LLM-based ENA model. In reference, in order to learn the LLM-based ENA model, unstructured data such as drawings should be generated as training data in the form of the sequence of tokens.

We demonstrate the effectiveness of the Earthwork Network Architecture (ENA) in automating and improving the accuracy of earthwork volume estimation through a case study and compare the performance of the ENA models developed using the MLP, LSTM, Transformers, and LLM to provide insights and challenges.

3. Framework for Automated Earthwork Network Architecture (ENA)

3.1. Overview

The proposed framework (Figure 1) aims to provide automated earthwork quantity estimation with high accuracy rates. The framework comprises three key modules: Topological Feature Collector, ENA Model Development, and Performance Evaluation.

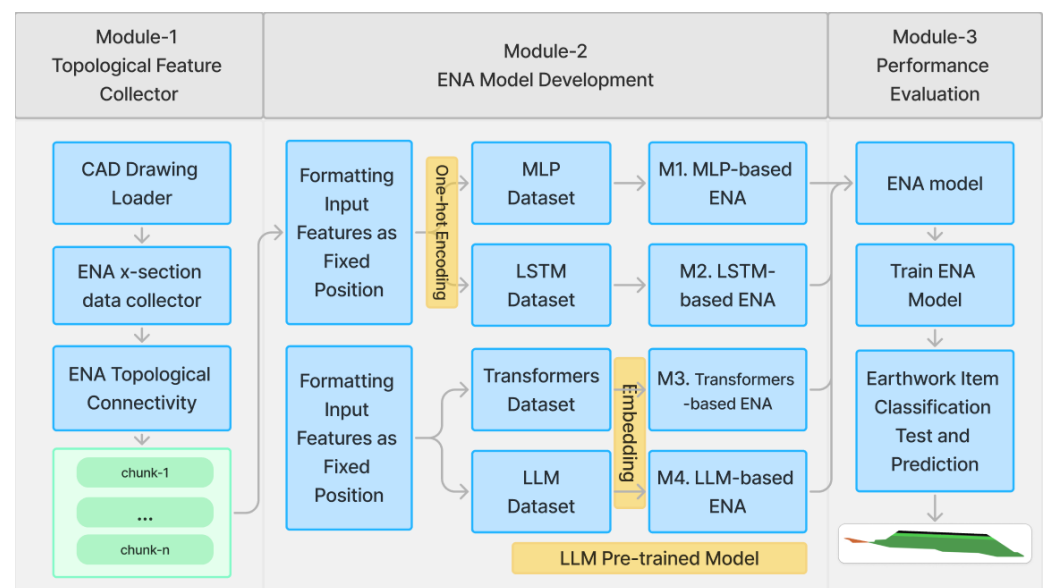


Figure 1. Framework for automated Earthwork Network Architecture.

Module-1 (Topological Feature Collector) extracts and organizes critical earthwork features from road cross-sectional designs or similar datasets. This module collects relevant topological data such as cut and fill areas, slope information, and subgrade conditions, converting them into a structured and trainable format. The data typically come from unstructured sources like CAD drawings, representing informal data commonly used in civil engineering. The key role of this module is to pre-process and tokenize the input data, transforming it into feature sequences that can be efficiently fed into machine learning models. This step is essential to ensure that the subsequent models can effectively learn from the topological features and make accurate predictions.

Module-2 (ENA Model Development) involves the design, training, and implementation of different deep learning models, including M1 (Multi-Layer Perceptron-based ENA), M2 (LSTM-based ENA), M3 (Transformers-based ENA), and M4 (LLM-based ENA) architectures. This module utilizes the pre-processed data from the Topological Feature Collector to train these models in predicting earthwork quantities. The primary objective here is to develop models that can accurately classify and estimate quantities from the topological features extracted. LLM-based models are particularly emphasized in this study as they are designed to handle sequences of tokens, making them highly effective for processing informal and unstructured data. Each model's architecture is fine-tuned to optimize performance, considering factors like loss reduction, accuracy, and convergence speed.

Module-3 (Performance Evaluation) assesses the accuracy and efficiency of the developed ENA models. In this phase, the models are tested on both training and test datasets to evaluate their ability to predict earthwork quantities. The evaluation metrics include loss (how well the model's predictions match actual values), Quantity Takeoff Classification accuracy (QTCacc), and model performance in terms of convergence speed. Additionally, the resource consumption (e.g., GPU usage) and computational efficiency of each model are considered. This module ensures that the developed models not only perform well on the training data but also generalize effectively to new, unseen datasets, making them applicable in real-world civil engineering scenarios. Special attention is given to comparing the performance of LLM-based models with traditional deep learning models like the MLP and LSTM, highlighting the advantages of using LLMs for complex, unstructured data. Figure 1 depicts the overall flow of the proposed framework, and detailed descriptions of each module are provided in the following section.

3.2. Module-1: Topological Feature Collector

Module-1: Topological Feature Collector in the proposed Earthwork Network Architecture (ENA) framework was responsible for the extraction, organization, and pre-processing of critical earthwork features from road cross-sectional designs or similar datasets. The module identifies and extracts important topological features such as cut and fill areas, slopes, subgrade conditions, and other geometric attributes from cross-sectional drawings. These features are crucial for accurate earthwork quantity estimation and serve as the primary inputs for the subsequent modeling process. In particular, the training data for LLMs should be in the form of sequences with some ordering among the features.

To extract the features required for training, it is necessary to create polylines by connecting line segments from entities included in the cross-section. Each entity in the cross-section is represented as a line segment on a specific layer, but these segments are not always topologically connected. To establish connectivity, the Half-Edge Topology Structure is introduced. This structure is used to trace the line in a specific direction, creating a closed polyline by ensuring that each segment is connected to form a complete entity.

The Half-Edge Topology Structure is an efficient method for representing the connectivity of the Boundary Representation (B-REP) model, which is widely used in computational geometry. It is particularly useful for managing complex operations such as geometry navigation and topological queries, making it an ideal choice for accurately modeling road cross-sections in CAD. Figure 2 illustrates a typical Half-Edge Topology Structure example.

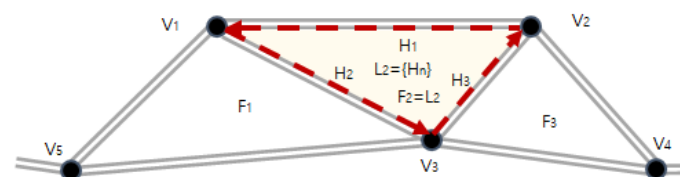


Figure 2. Half-Edge Topology Structure example.

The Half-Edge Topology Structure is composed of three main elements: V (Vertex), H (Half-Edge), and F (Face).

- **V (Vertex):** Each vertex in the entity is represented individually, storing attributes like position, color, and other geometric properties. Each vertex also maintains a reference to one of the half-edges that originates from it.
- **H (Half-Edge):** A half-edge represents one side of an edge in the B-REP model. In a full edge, there are two half-edges, each pointing in opposite directions. Each half-edge also stores references to its next half-edge in the clockwise direction around its loop, its opposite half-edge, the vertex it originates from, and the loop it belongs to.
- **F (Face):** A face is a polygonal region bounded by a loop of half-edges. Each face stores a reference to one of its bounding half-edges, making it easy to navigate and manage the connectivity of the surface geometry.

To create connected polylines from entities of the cross-sectional drawing, which have entities in CAD format (see Figure 3), the algorithm (see Figure 4) used is as follows:

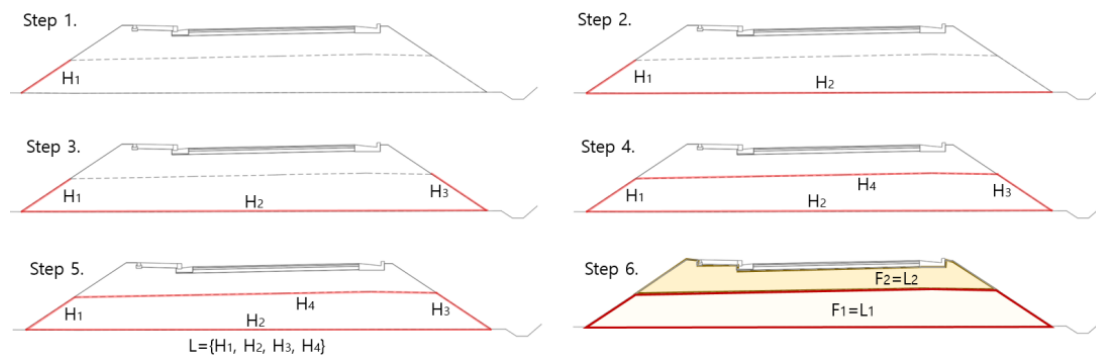


Figure 3. Example of creating a closed polyline.

T1. C_list = The entities stored in each cross-section are selected.
T2. E_list = Entities are selected for each layer in C_list . The entity is a line segment and consists of two H.
T3. H_seed = Randomly select H to track the phase in E_list .
T4. H_next = H_seed
T5. $L = \{H_next\}$
 T5.1. if $E_list = \{\}$ then break
 T5.2. $L = H_loop \cup H_next$
 T5.3. $E_list = E_list - H_next$
 T5.4. $H_next - list$ = Exploring other H lists with the same coordinate (x, y) value as the end point of the entity.
 T5.5. H_next = Select an entity with the largest internal angle between other entities and H_seed
 T5.6. if H_next in L then break
 T5.7. goto T5.1
T6. $P_list = P_list \cup L$
T7. $C_list = C_list - L$
T7. if $C_list = \{\}$ then exit
T8. goto T2

Figure 4. Pseudocode of creating a closed polyline using the Half-Edge Topology Structure and above algorithm.

Figure 4 illustrates the process of creating a closed polyline using the Half-Edge Topology Structure and the above algorithm. The steps depict how individual line segments are connected and looped to form a closed face, which is then used for further processing and analysis in the earthwork quantity estimation. This method provides a robust approach for establishing topological connectivity, making it easier to model complex cross-sectional geometries in CAD drawings.

After establishing connectivity, the module converts the geometric and topological data into a structured and trainable format. This includes tokenizing features, which is especially beneficial for models like Transformers and LLMs that require sequential data as input. The structured sequences enable the models to learn spatial relationships and dependencies more effectively, ensuring higher prediction accuracy.

The training feature vector f for each entity is standardized to include elements such as layer, length, and vertex coordinates:

$$\mathbf{vector}_{\text{data}} = \{\mathbf{Layer}, \mathbf{Length}, v\} \quad (1)$$

where

Layer: Includes attributes like name, line type, and color.

Length: Represents the geometric dimensions of the entity.

v : Denotes the x, y coordinates of the vertex.

Since entities in CAD drawings can vary significantly in representation, immutable elements such as layer names, vertical relationships, and intersection counts are used for consistency. The features are defined as:

$$f = \{\mathbf{Layer}_{\text{name}}, \mathbf{rel}_{\text{gnd}}, \mathbf{int}_{\text{pave}}\} \quad (2)$$

where

Layer_{name}: Captures the context of overlapping layers, such as ground, slope, and pavement layers.

rel_{gnd}: Describes the vertical relationship between the entity and the ground line, indicating whether the entity is above, below, or overlapping the ground line.

int_{pave}: Represents the number of intersections between the entity and pavement layers, crucial for area quantity characterization.

The feature rel_{gnd} determines whether an entity is above, below, or overlapping with the ground line, facilitating classification as a cut or fill. It is defined as

$$\mathbf{rel}_{\text{gnd}} = \begin{cases} \text{below} & \text{if } \max(D_y) < \min(G_y) \\ \text{above} & \text{if } \min(D_y) < \max(G_y) \\ \text{below} & \text{otherwise} \end{cases} \quad (3)$$

where

D_y is the y-coordinate of the design line;

G_y is the y-coordinate of the ground line.

Similarly, int_{pave} measures the intersections between the design lines and the pavement layers using

$$\mathbf{int}_{\text{pave}} = \{ \text{count}(ip) \mid ip \in IP \} \quad (4)$$

where

$IP = \{ x \mid \text{intersect}(V_{PB}, \text{pave}) \};$

$V_{PB} = (P_{BCP}, PL_{cp});$

P_{BCP} = center point of the pavement's bottom;

PL_{cp} = center point of each pavement layer;

$\text{pave} = \{\text{pavement}_{\text{layer}(i)} \mid i = \{1, \dots, n\}\};$

intersect = intersection calculation function between two lines.

These feature definitions help to distinguish between different entities and their relationships within a cross-sectional drawing, providing critical inputs for accurate earthwork.

The processed and structured data from the Topological Feature Collector become the primary input for the ENA models developed in Module-2. By organizing the data into a format suitable for model training, this module ensures that the deep learning models can accurately learn and generalize from the dataset, leading to better performance in Quantity Takeoff Classification and prediction tasks.

3.3. Module-2: ENA Model Development

In this module, various deep learning models were designed, trained, and implemented to predict earthwork quantities based on the pre-processed data from Module-1 (Topological Feature Collector). This module begins with the Feature Tokenizer Process, which is a critical step in converting input features from CAD drawings or other sources into a structured format suitable for deep learning models. The tokenizer is responsible for transforming each feature, such as spatial attributes and geometrical properties, into tokens that can be processed by models like Transformers and Large Language Models (LLMs).

Figure 5 defines the tokenization process centered on the training of Transformers-based ENA models (M3, M4). Note that other models, M1 and M2, differ in the one-hot encoding part and output.

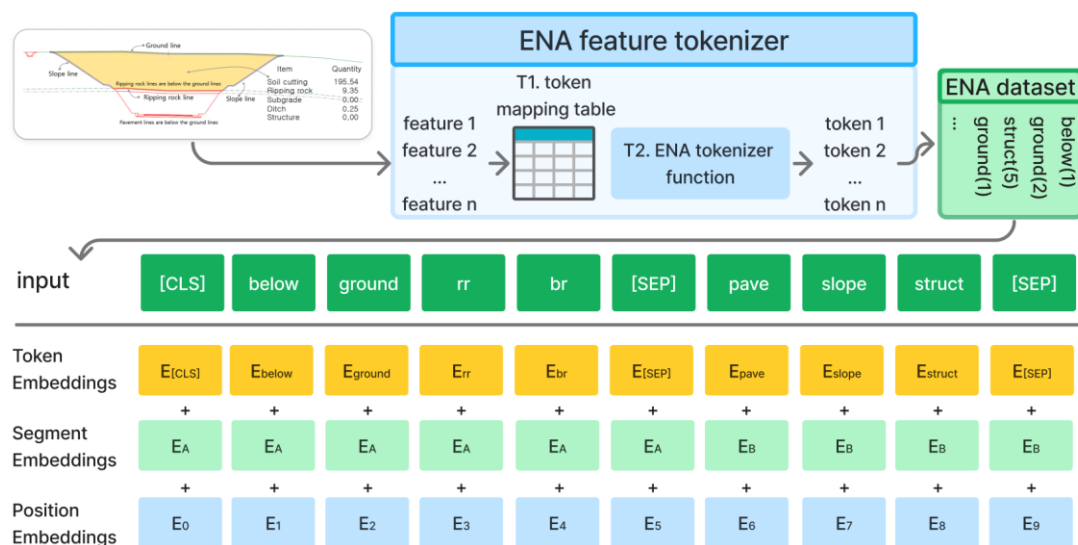


Figure 5. The process flow of the ENA Feature Tokenizer in Module-2.

It shows how these tokens were structured and embedded before being fed into the model, highlighting the multi-step transformation from raw input to a trainable format.

The process began with the extraction of key features from CAD drawings, including ground lines, slope lines, and structure layers, which were then mapped into a token table using a predefined token mapping strategy (T1). Each feature was assigned a unique token identifier, representing its position and role in the dataset. Following this, the ENA Feature Tokenizer function (T2) converted these mapped features into corresponding tokens, generating a sequence that encapsulated the spatial and topological features of the input data. This sequence was subsequently used to create the ENA dataset, which categorized earthwork-related attributes into various classes, such as cut areas, fill areas, and slope conditions.

Once tokenization was complete, the data were organized into a format that deep learning models could efficiently process. The figure illustrates how the data were represented using three main embedding components: token embeddings, segment embeddings, and position embeddings. Token embeddings captured the unique attributes and relationships of each feature in the dataset, translating the raw features into numerical vectors that the model could understand. Segment embeddings, on the other hand, provided additional context by indicating which part of the input data a particular token belonged to, helping the model interpret the grouping and segmentation of tokens within the dataset. Position embeddings specified the spatial position of each token within the sequence, ensuring that the model comprehended the order and positional relationships between features. This positional context was particularly critical for models like Transformers, which rely heavily on sequence information to learn complex patterns and dependencies.

The structured and tokenized representation of input data, as depicted in Figure 5, served as the foundation for training the ENA models. The figure demonstrates how this tokenization process enabled models to effectively learn from unstructured data by organizing and encoding the spatial and topological features into a cohesive format. This approach allowed the ENA framework to process intricate spatial relationships and geometrical attributes, thereby facilitating the development of models capable of making precise earthwork quantity estimations. The tokenization process enhanced the framework's ability to handle diverse and complex datasets within the construction domain, providing a robust and scalable solution for automating the estimation process. It illustrated the ENA Feature Tokenizer Process of this module, demonstrating the stages from feature formatting to model training. This workflow was integral to the ENA framework, which aims to automate and enhance earthwork quantity estimation using multiple deep learning architectures. Each component in the figure represents a step in the framework's systematic approach to structuring and processing data, followed by developing models based on the type of input data and desired outcomes.

Once the datasets were created, they were utilized to train four types of ENA models, each leveraging a unique deep learning architecture. The M1 model processed fixed-position data and was tailored for simpler datasets. The M2 model employed LSTM networks to learn temporal dependencies in sequential data, making it effective for applications that required time-based pattern recognition. The M3 model applied Transformer networks, which were designed to handle relationships between elements in large datasets, making them ideal for modeling complex dependencies. Finally, the M4 model integrated pre-trained BERT models, enabling it to process contextual information within sequence data. By leveraging BERT's pre-trained embeddings, the LLM-based model could learn from unstructured inputs with greater accuracy and efficiency.

The workflow culminated in the integration and training of these ENA models. Each model was fine-tuned to learn from its respective dataset, optimizing its performance based on the specific requirements of earthwork quantity estimation. The pre-trained BERT model, incorporated into the LLM-based ENA, further enhanced the model's ability to process and learn from complex datasets by providing contextual embeddings that added depth to the learning process.

The following equations define the ENA models M1 through M4.

M1. The MLP-based ENA was designed as (5) in consideration of training data structure and model tuning.

$$\begin{aligned}
 EAN_{MLP} &= \{front\ fc, MLP\ module, classification\ fc\} \\
 fc &= fully\ connection \\
 input &= input\ size \\
 hidden &= hidden\ layers's\ units\ size\ list \\
 output &= output\ size \\
 ReLU(x) &= \max(0, x) \\
 BatchNorm(x) &= (x - E[x]) / \sqrt{(Var[x] + \epsilon)} \times \gamma + \beta \\
 \gamma, \beta &= learnable\ parameter\ vectors\ of\ input\ size \\
 \hat{y} &= output\ of\ hidden\ layer \\
 \hat{y} &= predicted\ output
 \end{aligned} \tag{5}$$

M2. The LSTM-based ENA was designed as Equation (6) [40].

$$\begin{aligned}
 EAN_{LSTM} &= \{front\ LSTMs, MLP\ module, classification\ fc\} \\
 LSTMs &= \{LSTM*\} \\
 LSTM &= \{cell*\} \\
 cell &= \{i_t, f_t, o_t\}
 \end{aligned} \tag{6}$$

M3. The Transformers-based ENA was defined as Equation (7) [41] to train the ENA dataset.

$$\begin{aligned}
 EAN_{TR} &= \{front\ fc, TR_{encoder}, classification\ fc\} \\
 &here, \\
 TR_{enc} &= transformer_{encoder} \{feature, TR_{layer}^*\} \\
 EAN_{enc_layer} &= \{Embedding, Multihead\ Attention, Add\&Norm, MLP\} \\
 Multihead\ Attention &= \{Attention^*\} \\
 Attention &= softmax(QK^T / \sqrt{d_K}) \cdot V \\
 Add\&Norm &= \{addition, normalization\} \\
 Q &= query\ vector, K = key\ vector, V = value\ vector
 \end{aligned} \tag{7}$$

M4. The LLM-based ENA was defined as Equation (7). The model ran the base LLM after embedding the data to train the ENA dataset. The input sequence was converted to embedding vectors after tokenization.

$$\begin{aligned}
 EAN_{LLM} &= \{LLM_{pretrain}, fc, softmax\} \\
 &here, \\
 fc &= fully\ connected\ layer \\
 softmax(x_i) &= e^{x_i} / sum(e^{x_j}) \\
 e^{x_i} &= exponent\ of\ x_i \\
 e^{x_j} &= exponent\ of\ x_j \\
 sum &= summation\ function
 \end{aligned} \tag{8}$$

The LLM-based model training calculated the mask vector required for training in advance according to the length of the input sequence and was delivered to the model with the dataset. As a result, EAN_{LLM} was fine-tuned to suit the classification of earthwork.

Overall, Figure 5 encapsulates the progression from feature formatting to model development, showcasing how the ENA framework adapted to different data types and complexities. By employing various encoding techniques and model architectures, the framework presented a flexible and powerful solution for automating and refining earthwork quantity estimation. It demonstrated the adaptability of the ENA models to diverse input data, ensuring comprehensive and accurate predictions across a range of construction-related applications.

3.4. Module-3: Performance Evaluation

Module-3: Performance Evaluation in the ENA framework focused on assessing the accuracy, efficiency, and overall effectiveness of the developed models from Module-2. This module involved a series of structured activities that tested and validated the models' performances, ensuring that they could effectively predict and classify earthwork quantities under various conditions. The module began by consolidating all the trained models, including those based on the MLP, LSTM, Transformers, and LLM. These models were then subjected to rigorous training and evaluation processes to gauge their accuracy and reliability in real-world scenarios.

The initial step within this module was to train the ENA models using the prepared datasets. During this process, the models learned from the structured and pre-processed features extracted from CAD drawings or similar data sources. Each model architecture, whether it was MLP-, LSTM-, Transformer-, or LLM-based, was fine-tuned to optimize its learning capabilities. This stage involved iterating through multiple training epochs, adjusting hyperparameters, and refining the model's weights to ensure optimal learning from the input data.

Once the training phase was complete, the models proceeded to the test and prediction classification stage. In this phase, each model was evaluated using a designated test dataset that the model had not encountered during training. This testing phase was critical as it measured how well the model generalized to new, unseen data. Various evaluation metrics,

such as loss, accuracy, and QTCacc using Equation (9), were used to assess the models' performances. QTCacc, a specific metric developed for evaluating earthwork quantity estimation, was calculated by considering both the area and quantity item classification accuracies. This metric ensured that the models were not only predicting the correct quantities but also correctly classifying each quantity item within the dataset.

$$\begin{aligned}
 QTC_{acc} &= (P1 + P2)/2 \\
 P1 &= A1/(A1 + A2) \\
 P2 &= C1/(C1 + C2) \\
 A1 &= \text{Area of Ground True (m}^2\text{)} \\
 A2 &= \text{Area of Ground False (m}^2\text{)} \\
 Q1 &= \text{QTC item count of Ground True} \\
 Q2 &= \text{QTC item count of Ground False}
 \end{aligned} \tag{9}$$

During this evaluation phase, it was also essential to monitor the computational efficiency and resource consumption of each model, particularly for LLM-based models that tend to require significantly more computational resources. The LLM-based models, for example, might demonstrate higher accuracy and faster convergence rates but at the cost of increased GPU usage and longer training times. Understanding these trade-offs allowed for a more informed decision on which model architecture was best suited for a specific application or project constraint.

The final outcome of the Performance Evaluation module was a comprehensive comparison of the models' strengths and limitations. By analyzing their prediction accuracy, classification capabilities, and computational efficiency, this module provided insights into the suitability of each model architecture for automated earthwork quantity estimation. Additionally, it highlighted how advanced models like LLMs and Transformers can outperform traditional architectures when dealing with complex and unstructured datasets. This comprehensive evaluation is crucial in establishing the feasibility and effectiveness of using deep learning models for automated earthwork quantity estimation in the construction industry.

4. Performance Evaluation Through a Case Study

This study presents a case study to identify the performance of the ENA model proposed in the previous section. For this, the use case was selected from the earthwork quantity estimation work of a road construction project located in South Korea, and the prototype development of each module was developed. Then, the performance analysis was performed according to the ENA evaluation module.

4.1. Use Case Overview

The case study used for performance evaluation in this chapter involved a national road project with a total length of 5.46 km. The cross-sectional drawings for this project were specifically designed to represent various earthwork quantities and conditions encountered during the construction. A total of 237 cross-sections were utilized for training the model, incorporating 4051 distinct entities that express the earthwork quantities present within these cross-sections. Figure 6 shows several CAD cross-section drawings of the project that this study used.

4.2. Prototype Development

To compare the performance between the ENA models, an ENA prototype was developed (Figure 7). The models were derived from PyTorch Module, and forward () function of M1, M2, M3, and M4's ENA models were implemented. The architecture used in each model is the same as the formula described above. Autocad Python library was used to collect each entity's data in the cross-section. ENA_quantity_item, which expresses each earthwork item, kept the label and polyline. After generating the connectivity of the entity stored in the cross-section by layer, the features were calculated and they were stored in

the earthwork feature. Training features were converted into token sequences via one-hot encoding or the EAN_feature_tokenizer according to each model type. Transformers connected the Transformer Encoder output to the MLP to classify earthwork. The LLM used pre-trained BERT, and the same data format was used as a Transformers-based ENA model with a token sequence that was converted for model fine-tuning.

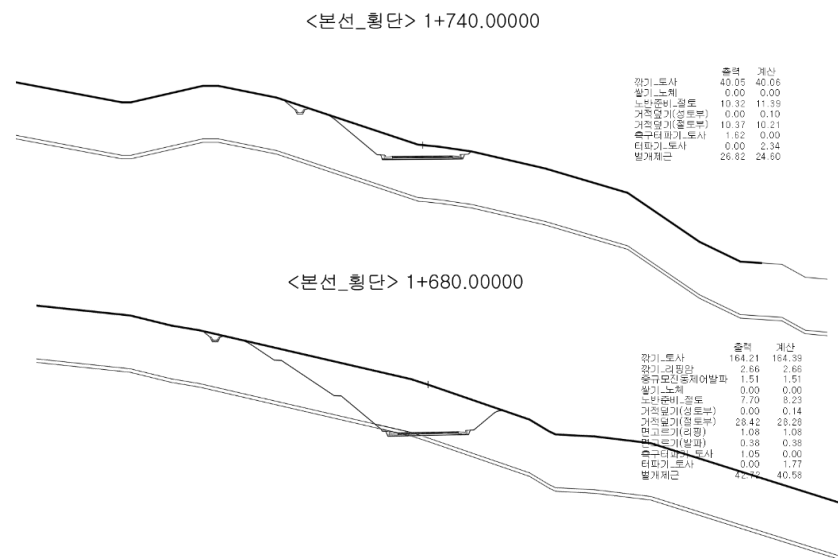


Figure 6. CAD drawings (a portion of the cross-sections) for the case study.

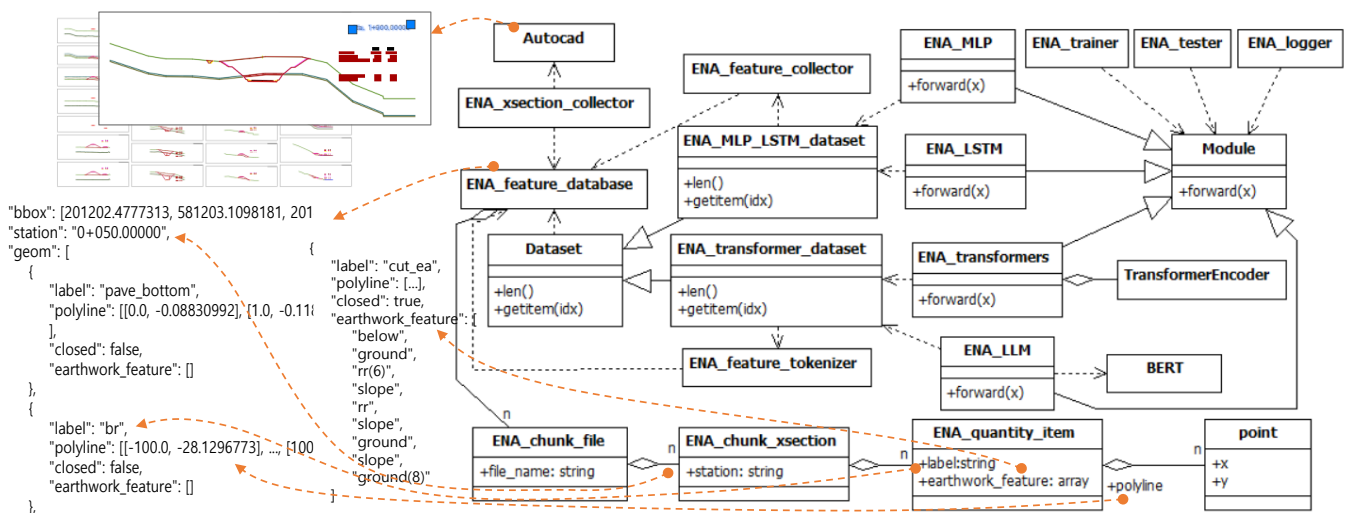


Figure 7. ENA prototype architecture (UML).

After the ENA's M1–4 models were developed in PyTorch (2.4), they were trained using hyperparameters like Table 1. The 524 GPU used was NVIDIA RTX 3080 with 8 GB RAM (Nvidia, Santa Clara, CA, USA). The table provides a comprehensive overview of the hyperparameters used to train various models within the ENA. These models were categorized into different architectures, including the MLP, LSTM networks, Transformers, and LLM, each with its own configuration and training setup.

For batch size, all MLP and LSTM models were trained using a batch size of 32, while the batch size for Transformers-based models varied across 32, 64, and 128, reflecting the increased computational demands of more complex architectures. In terms of epochs, MLP and LSTM models were trained for 150 epochs, ensuring adequate learning for these simpler models. However, for the more complex Transformers and LLMs, training was

extended to 300 epochs to allow the models sufficient time to learn from the data and converge on accurate predictions. The layer architecture column describes the specific configurations for each model.

Table 1. ENA model hyperparameters.

ENA Model ID	Batch Size	Epochs	Layer Architecture	Normalization and Activation	Learning Rate	Pre-Trained Model	
						Embedding	Transformers
M1.1. MLP	32	150	[14]-[128,64,32]-[10]	ReLU, batch normal, dropout	0.001	-	-
M1.2. MLP	32	150	[14]-[64,128,64]-[10]		0.001	-	-
M1.3. MLP	32	150	[14]-[64,128,64,32]-[10]		0.001	-	-
M1.4. MLP	32	150	[14]-[32,64,32]-[10]		0.001	-	-
M2.1. LSTM	32	150	[14]-LSTM[128]-[10]	Dropout	0.001	-	-
M2.2. LSTM	32	150	[14]-LSTM[128]-[64,32]-[10]		0.001	-	-
M2.3. LSTM	32	150	[14]-LSTM[256]-[128,64]-[10]		0.001	-	-
M3.1. Transformers	32	300	[320,512]-ENC[512]-[64]-[10]	Multi-head attention, layer normal, dropout	1.00×10^{-5}	BERT-base-uncased	-
M3.2. Transformers	64	300	[320,512]-ENC[512]-[64]-[10]		1.00×10^{-5}	BERT-base-uncased	-
M3.3. Transformers	128	300	[320,512]-ENC[512]-[64]-[10]		1.00×10^{-5}	BERT-base-uncased	-
M4.1. LLM	32	150	[293]-EMB-ENC-[10]		1.00×10^{-5}	BERT-base-uncased	BERT
M4.2 LLM	32	300	[293]-EMB-ENC-[10]		1.00×10^{-5}	BERT-base-uncased	BERT

The MLP models used fully connected layers with various configurations, such as [14]-[128,64,32]-[10], indicating the number of neurons in each layer. The LSTM models incorporated stacked LSTM layers with configurations like [14]-LSTM[128]-[10]. Transformers and LLMs, being more advanced, employed an embedding layer (EMB) followed by encoder layers (ENC) to process input data more effectively. Regarding normalization and activation functions, the MLP models rely on ReLU activations, batch normalization, and dropout for regularization. The LSTM models primarily use dropout to prevent overfitting, while Transformer models utilize multi-head attention, layer normalization, and dropout to enhance performance and ensure stability during training.

The learning rate was set at 0.001 for the MLP and LSTM models, which is a typical choice for these simpler architectures. For the more complex Transformers and LLMs, the learning rate was reduced to 1.00×10^{-5} to facilitate finer adjustments to the model parameters during training, avoiding overshooting and ensuring smooth convergence. Lastly, pre-trained models were used for the Transformers and LLM architectures. Specifically, these models leveraged BERT (base-uncased) embeddings to enhance their performance in processing text-based data for earthwork quantity estimation tasks. The use of pre-trained models allowed the network to start with more informed weights, reducing the amount of data needed for training and improving accuracy.

4.3. Training and Testing Results

Model training results are shown in Table 2. The loss had good performance for the LLM, LSTM, and MLP-based ENA models in the order of M4.2 (0.0534), M4.1 (0.0587), M2.2 (0.0851), and M1.2 (0.0852). The difference in loss improvement between M4.2 and M2.2 models was 0.0317, which shows a relatively large difference considering that the

average loss improvement value between models except M1.4 and M3.1 ~ 3, which had very poor performance, was 0.0051. The LLM-based model was fast at converging the solution. For reference, at epoch = 10, the loss of the MLP-based M1.1 model was 0.1685, but the loss was 0.074 for M4.1. The LLM-based ENA model showed a quick and stable exploration of optimal solutions. The prediction accuracy for the test dataset was in the order of M1.3 (0.9544), M4.2 (0.9531), M1.2 (0.9519), and M4.1 (0.9507). For reference, the Transformers-based ENA model tended to converge very late for training. The pre-trained LLMs had good training performances. However, the LLM-based model used many GPU resources (M4 model size = 427,783 KB), and the calculation speed was slow (M4 model training time = average 5 h 35 min).

Table 2. ENA models' performances.

ENA Model ID	Train No	Loss	Accuracy	Model Size (kb)	Time Performance (min)
M1.1. MLP	1650	0.0870	0.9494	61	0:02:34
M1.2. MLP	1714	0.0852	0.9519	84	0:02:36
M1.3. MLP	1716	0.0882	0.9544	93	0:02:42
M1.4. MLP	1718	0.1322	0.9507	30	0:02:25
M2.1. LSTM	1730	0.0889	0.9408	812	0:02:13
M2.2. LSTM	1732	0.0851	0.9420	850	0:02:17
M2.3. LSTM	1734	0.0886	0.9408	3312	0:02:00
M3.1. Transformers	2003	0.3533	0.7744	74,557	0:11:06
M3.2. Transformers	2014	0.3551	0.7719	74,557	0:07:48
M3.3. Transformers	2021	0.3596	0.7423	74,557	0:06:25
M4.1. LLM	0103	0.0587	0.9507	427,783	3:12:05
M4.2. LLM	2334	0.0534	0.9531	427,783	7:59:00

The loss and accuracy of the MLP-based model showed high performance under the GPU resources used, and the pre-trained LLM-based models showed fast solution exploration.

In Figure 8, the earthwork classification result of M3.2 is incorrectly predicted because it contains the bottom of the pavement layer. In contrast, M4.1 is under the ground line and has a sequence containing a part of the ground and the structure line, so it can be seen that the results are well predicted. Figure 9 shows some of the predictions for input sequences for each model. Here, the model output $\hat{y}$ of each sequence is displayed on the left side of the sequence bar. The right side shows the legend of the earthworks and earth lines marked by color.

4.4. Quantity Takeoff Classification (QTC) Performance Results

The ENA model's QTC performance comparison considered 3 + 600 to 4 + 600 stations, a crossing section containing various cross-sections in the dataset. A performance evaluation was performed on the earthwork quantity items included here (811 items in total) (Table 3). The QTCacc performance of the models was high in the following order (QTCacc Rank): LLM (M4.1 = 97.17%, M4.2 = 96.74%), LSTM (M2.2 = 82.10%), and MLP (M1.3 = 81.61%).

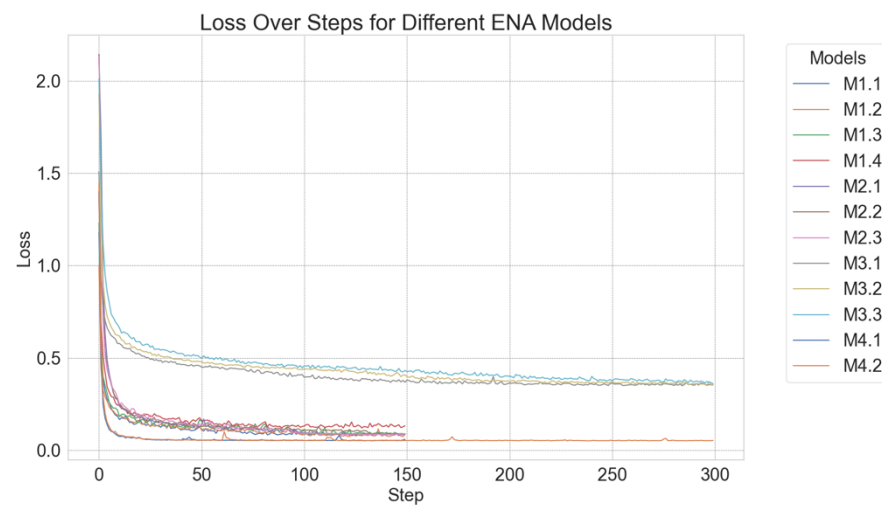


Figure 8. ENA models loss graph (x = epoch, y = loss).

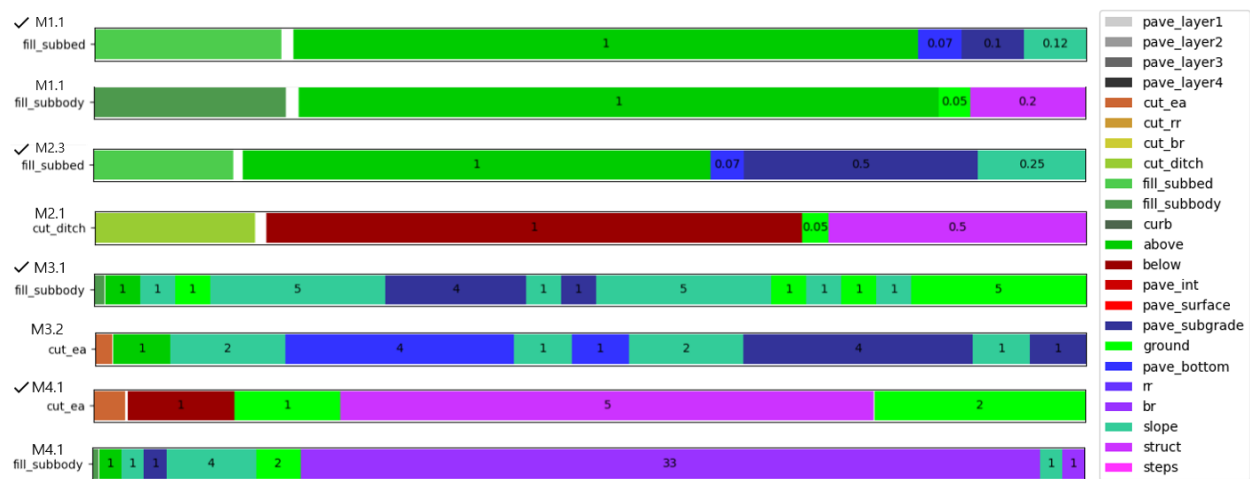


Figure 9. ENA model's prediction test results (data sequence bar. ✓: $y = \hat{y}$).

Table 3. ENA QTC performances (Area ratio = $T1/(T1 + F1)$. QTC ratio = $T2/(T2 + F2)$. ENA ratio = $(A1 + A2)/2$).

ENA Model ID	Ground True		Ground False		Accuracy			QTCacc Rank
	A1 (m ²)	C1	A2 (m ²)	C2	P1	P2	QTCacc (%)	
M1.1. MLP	9723.32	758	4300.33	53	69.34%	93.46%	81.40%	8
M1.2. MLP	9716.86	759	4306.78	52	69.29%	93.59%	81.44%	7
M1.3. MLP	9729.45	761	4294.20	50	69.38%	93.83%	81.61%	5
M1.4. MLP	9726.16	760	4297.48	51	69.36%	93.71%	81.54%	6
M2.1. LSTM	9762.06	762	4261.59	49	69.61%	93.96%	81.79%	4
M2.2. LSTM	9868.09	761	4155.55	50	70.37%	93.83%	82.10%	3
M2.3. LSTM	9512.40	756	4511.24	55	67.83%	93.22%	80.53%	9
M3.1. Transformers	847.35	133	13,176.29	678	6.04%	16.40%	11.22%	12
M3.2. Transformers	3347.50	40	10,676.14	771	23.87%	4.93%	14.40%	11
M3.3. Transformers	7317.13	73	6706.51	738	52.18%	9.00%	30.59%	10
M4.1. LLM	13,643.05	787	380.59	24	97.29%	97.04%	97.17%	1
M4.2. LLM	13,592.18	783	431.46	28	96.92%	96.55%	96.74%	2

To visually compare accuracy, the model prediction results for cross-sectional views showing large performance gaps between ENAs are shown in Figure 10. Looking at these results, it can be seen that there was a large difference in prediction accuracy between the M4.1 (LLM), M2.2 (LSTM), and M1.3 (MLP) models. An LLM-based model can better predict the area of earthworks. This shows that the LLM-based ENA, which is mainly used in NLP, can learn the earthwork quantity takeoff feature of the unstructured cross-sectional drawing dataset better than other models.

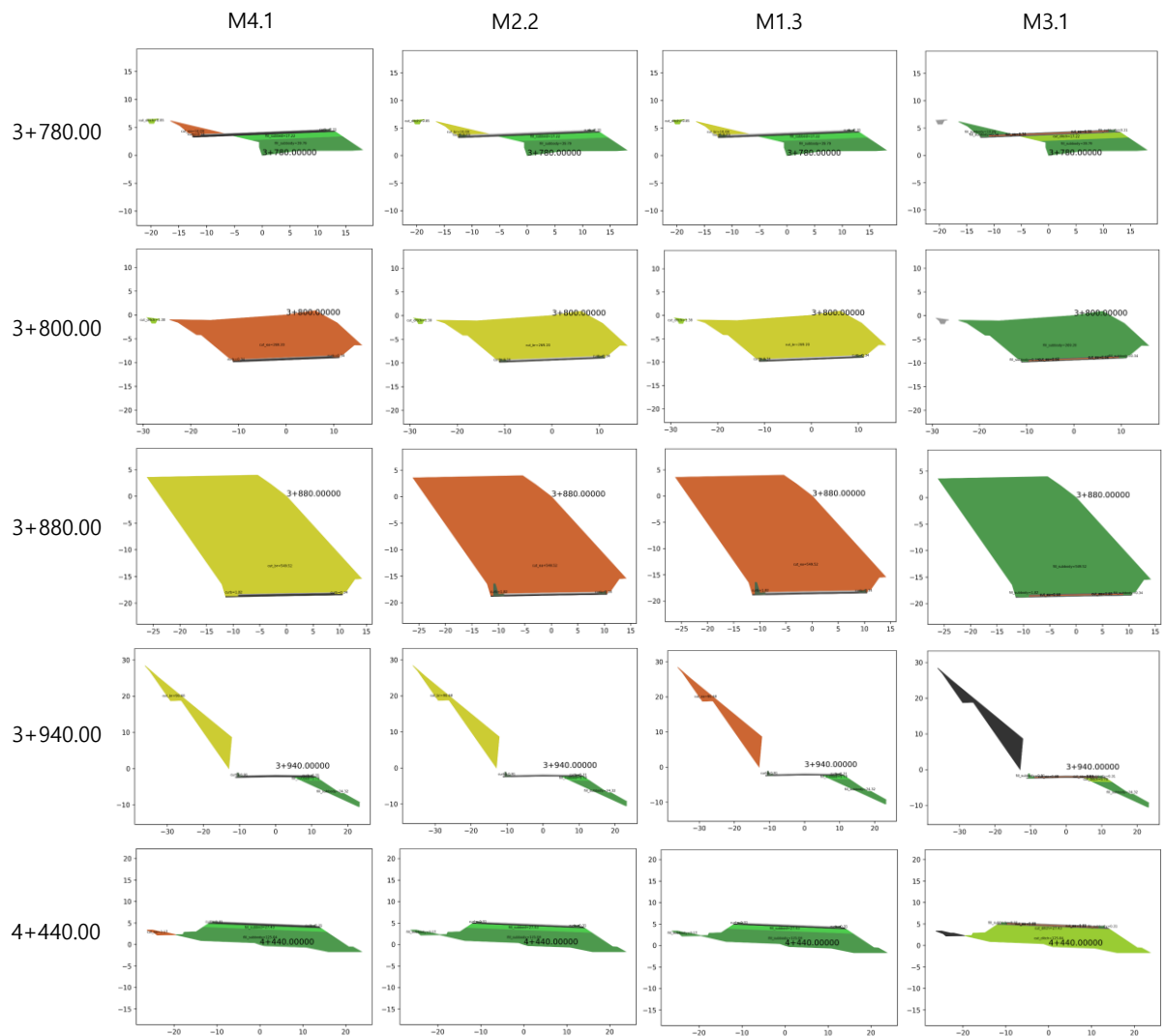


Figure 10. The ENA model's Earthwork Quantity Takeoff Classification estimation results (x: ENA model ID, y: stations. For the color code, refer to the earthwork item color code defined in Figure 9).

5. Discussion

In this study, key differences between the performance of the MLP, LSTM, and LLM-based ENA models for earthwork quantity classification are highlighted. This is different from previous studies [33,35] that train, classify, and predict the patterns through text sequences only in the AEC domain. Firstly, the MLP and LSTM-based models utilize one-hot encoding for training datasets, which, while effective, is limited in capturing complex relationships within the data. In contrast, the LLM-based ENA model employs feature tokens within the input sequence, allowing it to better understand and process unstructured data, such as CAD entities, by transforming them into a structured sequence of tokens. On the other hand, existing studies [28,39] using sequence modeling such as LSTM only implemented cases where the learning data were already expressed in the form

of time series dataset, so the research that considers such an informal dataset in AEC is not discussed.

Secondly, the LLM-based model excels at training sequence relationships within input tokens, unlike the MLP and LSTM models, which treat input variables more like regression curve fitting. This enables the LLM model to handle a broader range of data types, such as CAD drawings and images, and provides faster convergence. However, the LLM requires considerably more GPU resources. For instance, by epoch 10, the loss for the MLP-based M1.1 model was 0.1685, whereas the LLM-based model had already achieved a much lower loss. This faster convergence highlights the LLM's efficiency in managing complex datasets.

Thirdly, MLP and LSTM models are effective alternatives in environments where GPU resources are limited. Despite their more straightforward architectures, these models achieved high accuracy, between 0.94 and 0.95, with low loss values (0.08 to 0.13), and maintained smaller model sizes (84 KB to 3312 KB). This makes them a practical choice when computational resources are scarce, though their ability to handle more complex data is comparatively limited.

Fourthly, the LLM-based ENA model demonstrated superior performance in handling unstructured vector data, such as that used in earthwork tasks. Its training accuracy reached 0.95, with an exceptional QTCacc of 97.17%. This shows the LLM model's ability to encode and process diverse input features, making it highly versatile for various construction-related applications. In addition, there was also the previous study using Transformers [29], but since the training dataset was developed based on virtual scenarios rather than actual data in AEC, there was no realistic discussion.

However, adopting LLM-based models requires careful consideration. The first critical factor is the development of an effective strategy for preparing training data. Datasets like CAD drawings must be tokenized into sequences, and the way these sequences are encoded can significantly impact the model's training performance and efficiency. Optimizing the encoding process is essential to fully leverage the LLM's capabilities. Second, while Transformer-based models reduce the need for extensive feature engineering, encoding training data for the LLM-based ENA model still requires considerable effort to ensure that the token sequences are well structured. Additionally, due to the LLM's high demand for GPU resources, it is crucial to optimize the encoding process to minimize computational costs. Lastly, training Transformer models with small datasets often leads to suboptimal performance, as demonstrated by the M3.1 model, which only achieved an accuracy of 0.774. To improve performance when dealing with limited data, it is advisable to use pre-trained models or employ data augmentation techniques such as Retrieval-Augmented Generation (RAG).

While LLM-based ENA models offer powerful solutions for handling complex and unstructured data, their implementation requires a thoughtful approach to data encoding, resource management, and training data availability. With proper optimization, these models can significantly enhance the accuracy and efficiency of earthwork quantity estimation in the construction industry.

6. Conclusions

The rapid development of LLM (Large Language Model) technology, which excels at converting multimodal data formats in latent space, offers significant potential for application in the AEC (Architecture, Engineering, and Construction) industry. This is particularly valuable in a field that frequently deals with informal and unstructured data, such as CAD drawings and complex earthwork features.

This study aimed to develop an ENA model for automatic classification of earthwork quantity, design the model learning strategy for it, and evaluate its performance. Specifically, we extracted earthwork features from the road cross-section designed in CAD format. These were examples of unstructured data, and these features were trained using various deep learning models, including MLP-, LSTM-, Transformers-, and LLM-based ENA models.

As expected from the hypothesis, the results demonstrate that pre-trained LLMs exhibited superior performance, achieving better loss values compared with traditional deep learning models during earthwork classification tasks. For instance, the LLM-based model showed a significant improvement in convergence speed, with a loss of 0.1685 by epoch 10, outperforming other models. However, it is important to note that LLM-based models require substantial GPU resources (at least $129\times$ more GPU RAM than other models), emphasizing the need to select the appropriate model based on the use case and available computational environment.

Through this study, we identified several key advantages of the LLM-based ENA model. One of the primary benefits is its ability to handle multimodal data types through a simpler input sequence format, which enables more efficient feature encoding of unstructured data, such as CAD entities. This feature positions LLMs as a transformative technology with the potential to significantly advance AI applications in the AEC industry, particularly in scenarios involving informal and multimodal datasets.

Although Transformer-based LLMs are already recognized as a cornerstone for generating AI solutions, their application in the AEC domain is still in its infancy. In AEC, graph data such as CAD drawings and Industrial Foundation Classes (IFCs) are frequently used in decision-making processes. Looking ahead, there is substantial opportunity to expand on this study by exploring more effective embedding techniques, refining LLM architectures, and developing generative AI systems that support multimodal data processing within the AEC industry. As LLM technology continues to evolve, it will likely play an increasingly pivotal role in addressing the unique challenges of handling unstructured data in construction and engineering applications.

In the future, we plan to further develop cases and methods in which generative AI technologies such as LLMs can be used in various construction fields, and to develop them into multi-agent systems in AEC.

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